Supplementary Material for "Beyond the Random Walk: Horizon-Weighted Temporal Fusion Transformers for Thai Agricultural Commodity Price Forecasting"

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S1 Fairness protocol for reference-model configuration

A recurring concern in comparative forecasting studies is that the proposed model receives careful tuning while reference models run at defaults, biasing the comparison. This supplement documents the configuration provenance of every learned model in the study and reports a hyperparameter search for each learned reference class of Table 5 under one common protocol.

- **Matched architecture-search budgets.** The TFT architecture was informed by a 10-trial exploratory search (Section S2) and held constant across all seventeen loss exponents. Each learned reference class receives the same 10-configuration architecture or hyperparameter budget (Section S3), and the configuration used in the main paper is always one candidate. The separate seventeen-point loss sweep is the paper's primary method ablation and is disclosed in full; the total tuning budgets are therefore not identical.
- **Validation-only selection.** Every candidate is fitted on labels within 2018–2022 only and scored by matched-window MAE on validation windows whose horizon-h label falls inside 2023 (12,120 windows at t+20, 28,280 at t+60, 52,520 at t+120, and 105,040 at t+250 over the 404 crops), averaged over $h \in \{20, 60, 120, 250\}$. No test-period information enters any selection.
- **Identical final conditions.** The validation-selected configuration is then refitted on labels through 2023, the same training window the published reference models used, and evaluated on the identical 2024–2025 matched test windows as Table 5 of the main paper (110,292 windows per horizon, 404 crops).

S2 TFT architecture provenance

The TFT architecture used throughout the main paper (hidden size 64, 4 attention heads, continuous-variable encoding size 8, dropout 0.1, learning rate 0.01) was fixed before the decay-exponent sweep and was informed by a 10-trial exploratory Optuna search (median pruner, 3-epoch budget per trial) under the unweighted quantile objective. Table S1 lists the search space. The historical trial ledger was not retained, so this is architecture provenance rather than a reproducible claim that these values were the unique validation optimum. The architecture was then held constant across all seventeen decay exponents of the main experiment. The exponent was swept exhaustively over $\gamma \in \{0.0, 0.5, \ldots, 8.0\}$, with full per-horizon results disclosed in Table 6 of the main paper.

Table S1: Search space of the 10-trial TFT architecture search.

Hyperparameter	Space
Hidden state size	$\{16, 32, 64\}$
Attention heads	$\{1, 2, 4\}$
Dropout	$[0.1, 0.3]$
Continuous-variable encoding size	$\{8, 16, 32\}$
Learning rate	log-uniform $[10^{-3}, 10^{-1}]$

S3 Reference-model search spaces and validation results

Tables S2–S5 report all ten candidate configurations per class with their per-horizon validation MAE. Configuration 0 is always the original configuration used in the main paper. The gradient-boosting search varies tree complexity, regularization, and sampling; the MLP search varies width, dropout, and learning rate around the original values; the LSTM search varies hidden-state size and learning rate; the Transformer search varies model width, depth, head count, and learning rate. Training budgets (boosting rounds, epochs, batch sizes) are held at their original values within each class so that the search isolates configuration rather than compute. The GRU variant of the recurrent class is omitted from the search for the reason given in Section 4.3 of the main paper; it is not part of Table 5 and does not outperform the naive baseline overall.

Table S2: Global LightGBM search. Validation MAE (THB) per horizon; the validation-selected configuration is bolded.

Cfg	Parameters	t+20	t+60	t+120	t+250	Mean
0 (original)	leaves 2633, depth 8, min/leaf 700, lr 0.05, ff 0.99, bf 0.74, freq 3	26.79	46.95	62.81	77.03	53.39
1	leaves 31, depth -1, min/leaf 20, lr 0.05, ff 1.0, bf 1.0, freq 0	28.98	47.03	64.49	76.29	54.20
2	leaves 63, depth -1, min/leaf 20, lr 0.1, ff 0.9, bf 0.8, freq 1	30.52	47.34	63.65	76.44	54.49
3	leaves 127, depth 8, min/leaf 100, lr 0.05, ff 0.9, bf 0.8, freq 1	32.21	48.21	64.02	75.53	54.99
4	leaves 255, depth 10, min/leaf 100, lr 0.05, ff 0.8, bf 0.8, freq 1	32.31	46.92	63.64	75.35	54.56
5	leaves 511, depth 10, min/leaf 300, lr 0.05, ff 0.9, bf 0.8, freq 1	27.94	47.60	62.64	74.51	53.17
6	leaves 1023, depth 12, min/leaf 300, lr 0.03, ff 0.9, bf 0.7, freq 3	38.98	55.91	72.21	82.99	62.52
7	leaves 2633, depth 8, min/leaf 700, lr 0.1, ff 0.99, bf 0.74, freq 3	26.19	47.80	63.07	76.30	53.34
8 (selected)	leaves 4095, depth 12, min/leaf 1000, lr 0.05, ff 0.8, bf 0.7, freq 3	**24.62**	**44.76**	**62.52**	**74.63**	**51.63**
9	leaves 255, depth -1, min/leaf 50, lr 0.1, ff 1.0, bf 0.9, freq 1	36.40	45.94	64.89	77.09	56.08

Table S3: Global MLP search. Validation MAE (THB) per horizon; the validation-selected configuration is bolded.

Cfg	Parameters	t+20	t+60	t+120	t+250	Mean
0 (original)	h_1 128, h_2 64, dropout 0.47, lr 0.0034	42.51	53.62	63.96	84.44	61.13
1	h_1 128, h_2 64, dropout 0.1, lr 0.001	34.18	49.51	64.93	94.60	60.81
2	h_1 128, h_2 64, dropout 0.3, lr 0.003	33.83	45.31	58.24	78.03	53.85
3	h_1 256, h_2 128, dropout 0.3, lr 0.001	35.25	50.47	64.86	89.54	60.03
4	h_1 256, h_2 128, dropout 0.47, lr 0.0034	41.24	51.93	65.94	87.68	61.70
5 (selected)	h_1 64, h_2 32, dropout 0.1, lr 0.01	**32.33**	**43.66**	**56.07**	**79.75**	**52.95**
6	h_1 512, h_2 256, dropout 0.3, lr 0.001	33.68	50.53	69.46	102.13	63.95
7	h_1 128, h_2 64, dropout 0.47, lr 0.01	56.20	66.68	73.63	86.47	70.74
8	h_1 256, h_2 128, dropout 0.1, lr 0.003	29.25	47.10	67.98	95.15	59.87
9	h_1 64, h_2 32, dropout 0.3, lr 0.0034	39.83	52.14	68.62	90.19	62.69

Table S4: Global LSTM search. Validation MAE (THB) per horizon; the validation-selected configuration is bolded.

Cfg	Parameters	t+20	t+60	t+120	t+250	Mean
0 (original)	hidden 16, lr 0.01	51.87	59.75	69.26	80.11	65.25
1 (selected)	hidden 32, lr 0.01	**43.27**	**55.61**	**65.91**	**81.46**	**61.56**
2	hidden 64, lr 0.01	39.87	52.50	69.52	95.43	64.33
3	hidden 128, lr 0.01	38.33	52.05	68.84	93.43	63.16
4	hidden 16, lr 0.003	71.39	81.18	90.72	102.56	86.46
5	hidden 32, lr 0.003	50.91	61.70	71.17	89.97	68.44
6	hidden 64, lr 0.003	44.49	56.74	67.95	90.85	65.01
7	hidden 128, lr 0.003	41.30	54.90	69.00	96.78	65.50
8	hidden 64, lr 0.001	52.54	64.88	72.75	92.78	70.74
9	hidden 128, lr 0.001	42.90	53.54	66.06	93.83	64.08

Table S5: Global Transformer search. Validation MAE (THB) per horizon; the validation-selected configuration is bolded.

Cfg	Parameters	t+20	t+60	t+120	t+250	Mean
0 (original)	d_{model} 32, layers 2, heads 4, lr 0.003	60.08	66.45	70.42	104.78	75.43
1	d_{model} 64, layers 2, heads 4, lr 0.003	49.37	56.98	64.30	92.84	65.87
2	d_{model} 64, layers 4, heads 4, lr 0.001	48.98	57.98	62.67	89.79	64.85
3	d_{model} 32, layers 4, heads 4, lr 0.003	50.67	55.67	63.59	87.13	64.27
4 (selected)	d_{model} 64, layers 2, heads 8, lr 0.001	**42.38**	**53.48**	**66.15**	**90.44**	**63.11**
5	d_{model} 128, layers 2, heads 4, lr 0.001	50.13	62.69	71.52	97.24	70.39
6	d_{model} 32, layers 2, heads 4, lr 0.01	54.09	53.46	72.00	109.81	72.34
7	d_{model} 64, layers 4, heads 8, lr 0.0003	59.86	66.59	76.77	102.02	76.31
8	d_{model} 128, layers 4, heads 8, lr 0.001	42.20	54.14	69.24	89.25	63.71
9	d_{model} 32, layers 2, heads 2, lr 0.001	45.07	53.06	64.25	94.62	64.25

S4 Test-set impact of validation-selected configurations

Table S6 reports the 2024–2025 matched-window test MAE of each validation-selected configuration after refitting on labels through 2023. The "original (rerun)" column refits configuration 0 inside this supplement's pipeline under the same random initialization as the search candidates, and the "main paper" column repeats the corresponding overall value of Table 5. For the gradient-boosting class the rerun reproduces the main-paper value essentially exactly, which validates the pipeline; for the neural classes the rerun differs from the main-paper value by several THB in either direction under an identical configuration, illustrating initialization variance. Tuning-attributable differences should therefore be read against the rerun column rather than the main-paper column. The matched Lag-1 persistence baseline scores 45.21 THB on the same windows.

Table S6: Test MAE (THB) of validation-selected reference configurations against the original Table 5 configurations.

		Selected configuration					Original	Main
Model class	Cfg	t+20	t+60	t+120	t+250	Overall	(rerun)	paper
Global LightGBM	8	29.24	40.43	50.25	62.29	45.55	44.83	44.83
Global MLP	5	26.19	34.08	49.74	82.08	48.02	51.38	56.88
Global LSTM	1	41.12	45.94	54.59	76.79	54.61	54.58	59.59
Global Transformer	4	38.66	46.29	57.00	87.14	57.27	52.88	70.82

S5 Selected-exponent seed sensitivity

Table S7 reports independent retraining of the selected $\gamma = 4.5$ configuration. Exponent selection itself used the common seed-42 sweep and 2023 validation MAE; seeds 43 and 44 are post-selection sensitivity runs. All three runs beat persistence overall and at t+120 and t+250, while all remain worse at t+20. This supports the qualitative horizon pattern but also shows that the exact overall point estimate varies from 39.76 to 43.70 THB.

Table S7: MAE (THB) of the selected exponent under three independent seeds.

Seed	t+20	t+60	t+120	t+250	Overall
42	26.51	33.81	45.75	52.97	39.76
43	25.37	35.59	43.82	64.09	42.22
44	27.31	42.80	45.26	59.44	43.70
Persistence	15.11	30.86	51.03	83.84	45.21

S6 Data cleaning and exploratory diagnostics

Zero prices are treated as missing, rolling 30-day IQR outliers are removed, remaining gaps are forward-filled after business-day alignment, and a one-percent repricing threshold defines the 404-product modeling universe. Table S8 reports the raw inventory; Table S9 reports level-stationarity rates among the 635 active products. Figures S1–S4 provide the full diagnostics summarized in Section 3 of the main paper.

Table S8: Product counts and status by group.

Group	Active	Empty/all-zero	Total
Meat & Poultry	98	14	112
Aquatic & Marine	39	6	45
Fresh Vegetables	89	0	89
Organic Veg & Fruits	44	0	44
Fruits	64	8	72
Food Crops	82	11	93
Oil Crops & Oils	95	41	136
Wholesale Rice & Bags	54	0	54
Wholesale Rice to Retailers	10	0	10
Retail Rice	9	1	10
Field Crops	40	4	44
Feed & Raw Materials	11	4	15
Miscellaneous / Unknown	0	5	5
Total	**635**	**94**	**729**

S7 Detailed model and loss specification

All in-domain learned references use leakage-safe price lags, differences, returns, rolling statistics, repricing counters, calendar coordinates, and entity identifiers. LightGBM is direct by horizon;

Table S9: ADF level-stationarity profiles by market category ($p < 0.05$).

Category	Active	Minimum price	Maximum price
Retail	269	35.7%	33.1%
Wholesale	366	27.3%	25.7%

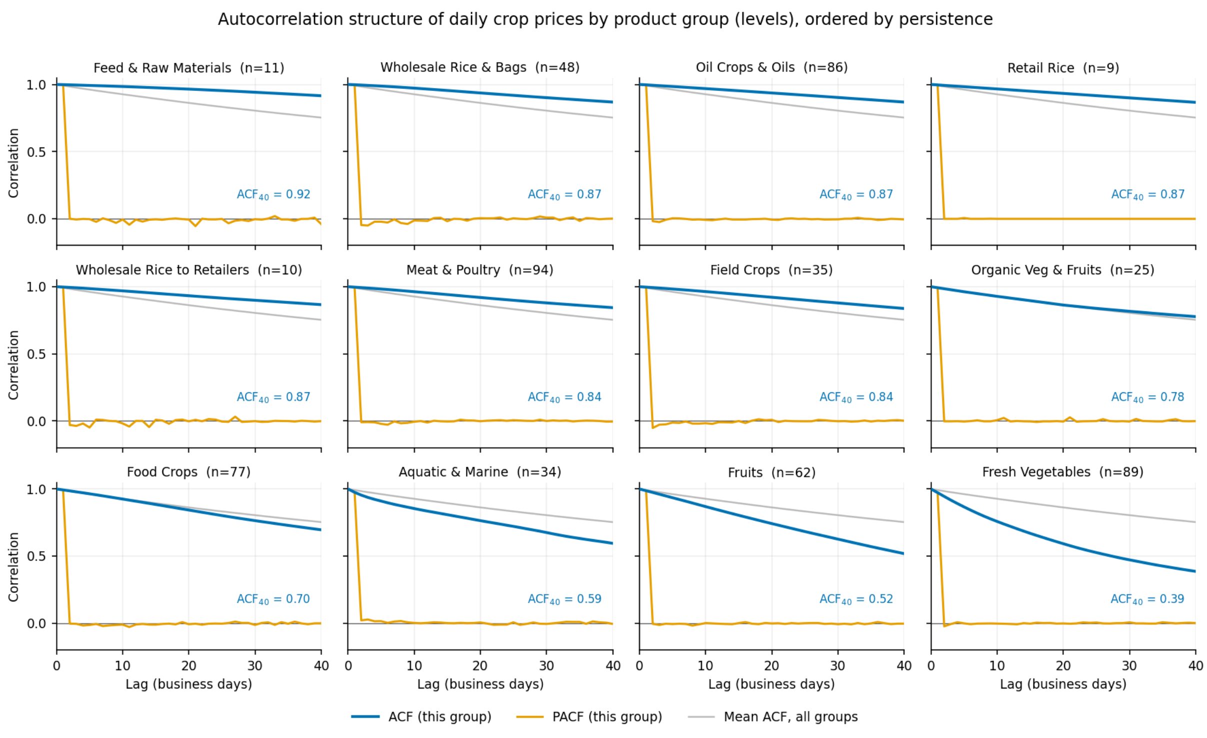

Figure S1: Mean ACF and PACF in levels by product group.

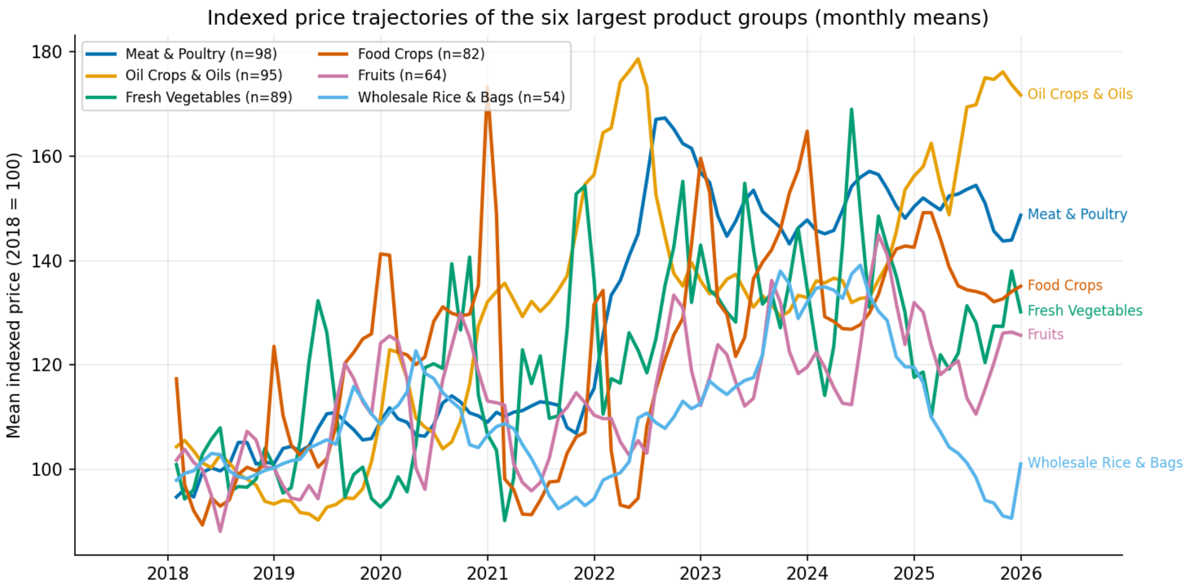

Figure S2: Indexed monthly price trajectories of the six largest groups.

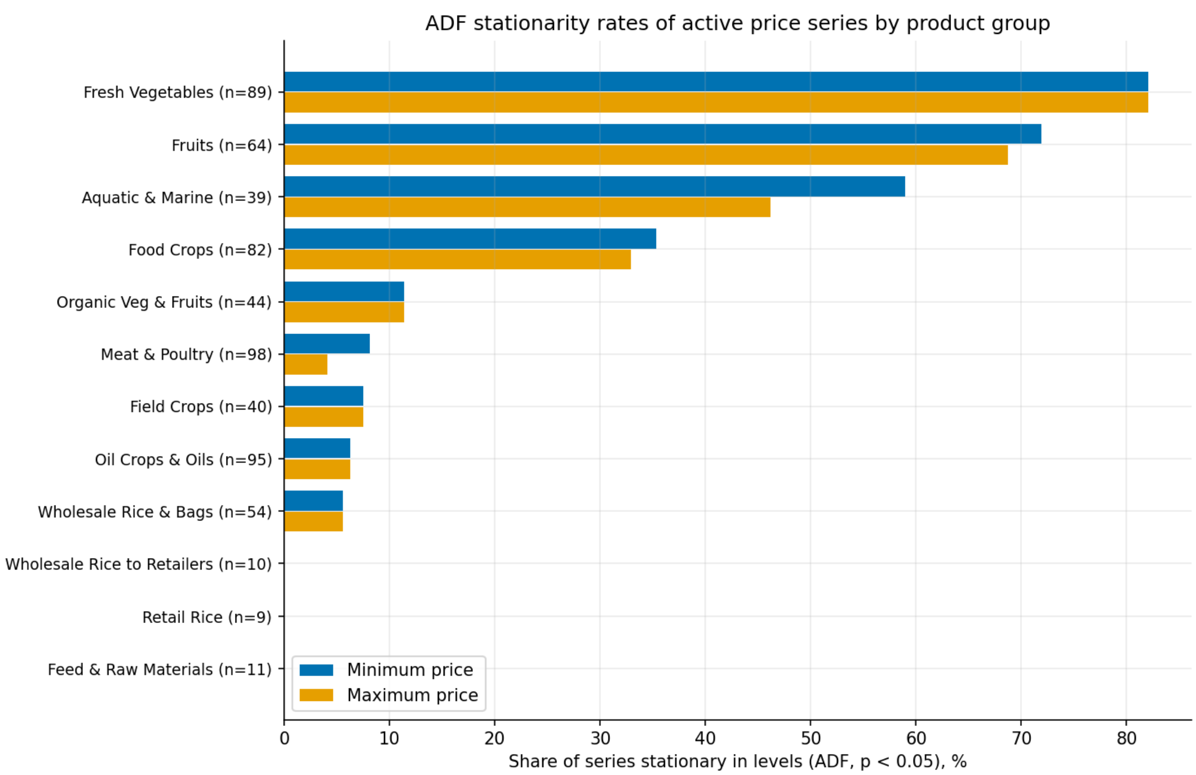

Figure S3: ADF stationarity rates by product group.

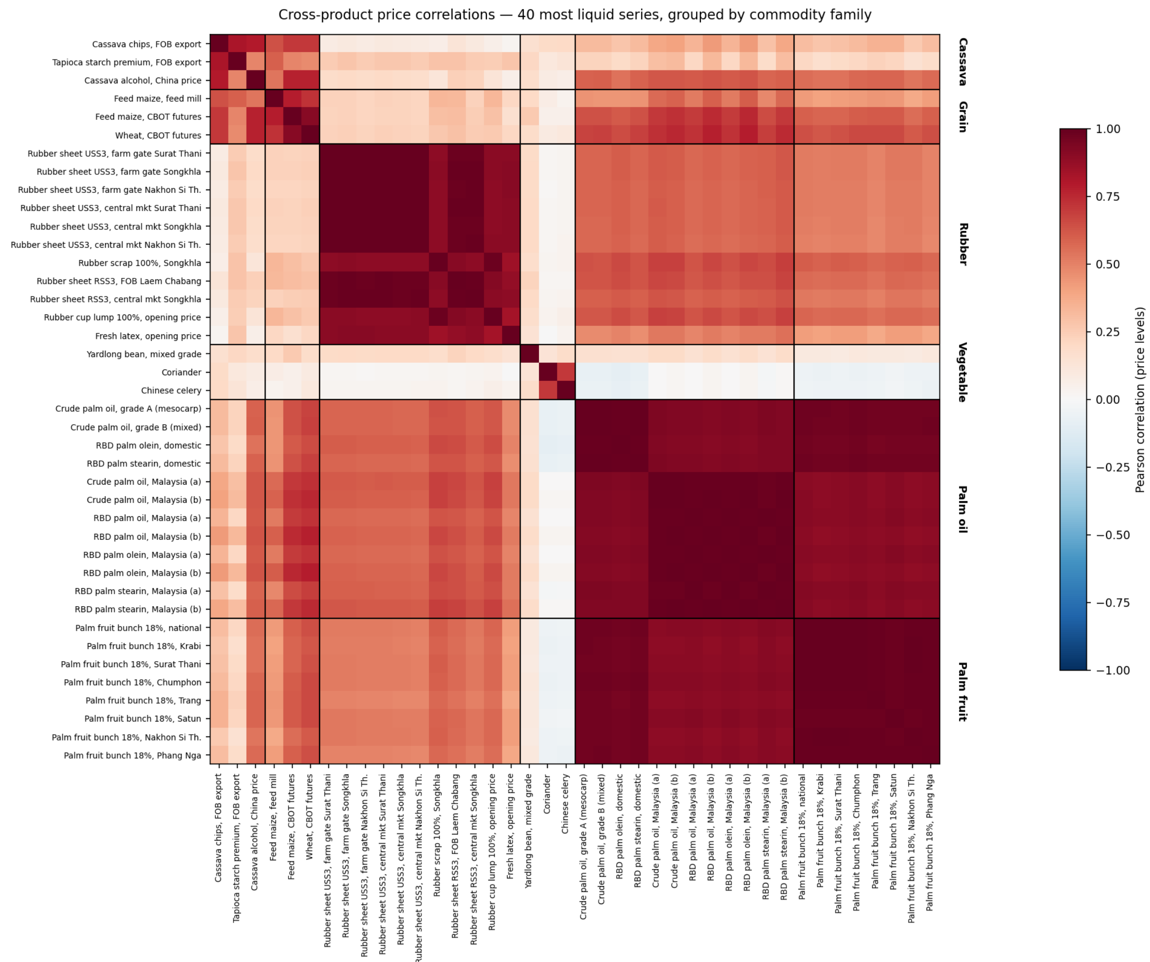

Figure S4: Price-level correlation among the 40 most liquid series; this deliberately integrated subset is not representative of the full universe, whose mean pairwise level correlation is 0.180.

the neural references use a joint multi-output head; ARIMA is local; and Chronos-Bolt-Base is zero-shot. The released source contains the exact feature list and configurations.

The TFT maps product, category, and group identifiers to static embeddings. Gated residual networks (GRNs) create context vectors for variable selection, LSTM initialization, and static enrichment. With an optional projected residual $\mathbf{W}_s\mathbf{a}$, a GRN is

$$\mathrm{GRN}(\mathbf{a}, \mathbf{c}) = \mathrm{LayerNorm}\{\mathbf{W}_s\mathbf{a} + \mathrm{GLU}(\mathbf{W}_1\,\mathrm{ELU}(\mathbf{W}_2\mathbf{a} + \mathbf{W}_3\mathbf{c} + \mathbf{b}_2) + \mathbf{b}_1)\}.$$

The variable-selection network combines independently transformed variables as

$$\tilde{\mathbf{x}}_t = \sum_{j=1}^{M} v_t^{(j)}\,\mathrm{GRN}_j(\mathbf{x}_t^{(j)}), \qquad \mathbf{v}_t = \mathrm{Softmax}(\mathrm{GRN}_v(\mathbf{g}_t, \mathbf{c})).$$

Interpretable multi-head attention uses head-specific query/key projections, a shared value projection, and an averaged attention matrix,

$$\tilde{\mathbf{A}} = \frac{1}{M}\sum_{m=1}^{M} \mathrm{Softmax}\left(\frac{(\mathbf{Q}\mathbf{W}_Q^{(m)})(\mathbf{K}\mathbf{W}_K^{(m)})^\top}{\sqrt{d_{\mathrm{attn}}}}\right), \quad \mathrm{IMHA} = \tilde{\mathbf{A}}\mathbf{V}\mathbf{W}_V\mathbf{W}_O.$$

The output heads target quantiles $\{0.1, 0.5, 0.9\}$ with pinball loss $\rho_q(e) = e(q - \mathbb{1}\{e < 0\})$. The proposed objective is

$$\mathcal{L} = \sum_{h=1}^{250} h^{-\gamma} \sum_{q\in\{0.1,0.5,0.9\}} \rho_q(y_{t+h} - \hat{y}_{t+h,q}).$$

Pinball-loss subgradients are bounded, so the paper interprets this as explicit loss allocation across decoder steps rather than as a direct measurement of gradient-vector conflict.

S8 Complete loss-exponent sweep

Table S10 reports the complete seed-42 grid used for validation selection; Figure S6 shows the corresponding test-MAE curves. Each exponent is one run, so local non-monotonicity should not be interpreted as a smooth dose response.

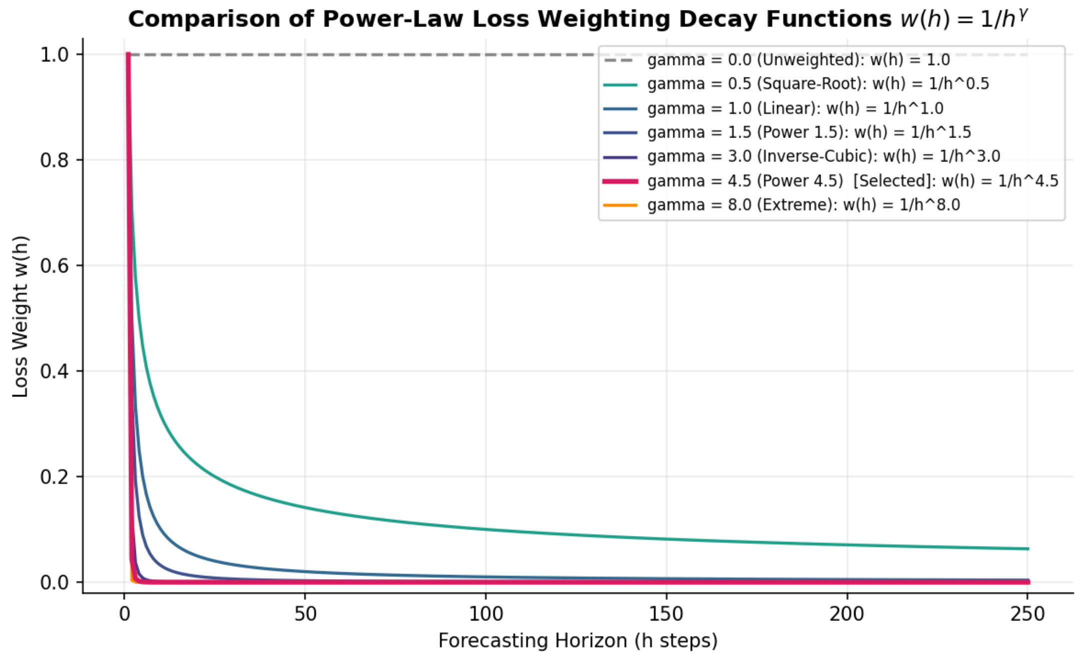

Figure S5: Power-law loss schedules over the 250-step decoder.

Table S10: Seed-42 horizon-weighted TFT results (MAE THB/SMAPE).

γ	t+20	t+60	t+120	t+250	Overall MAE
0.0	69.74/18.66%	76.37/21.77%	73.94/22.84%	71.37/23.21%	72.85
0.5	75.16/19.79%	78.42/23.03%	71.88/23.31%	63.94/23.49%	72.35
1.0	51.28/13.03%	64.83/19.53%	67.71/21.06%	72.03/23.40%	63.96
1.5	49.45/12.31%	72.24/20.76%	72.97/22.47%	71.21/22.31%	66.47
2.0	51.05/12.03%	62.63/18.40%	65.05/21.16%	62.45/22.25%	60.30
2.5	44.76/12.30%	59.10/19.88%	56.84/21.42%	60.30/21.93%	55.25
3.0	31.16/9.87%	38.24/15.98%	42.69/18.23%	57.63/18.55%	42.43
3.5	45.02/10.98%	50.74/18.27%	50.83/20.11%	47.92/18.77%	48.63
4.0	25.63/8.69%	34.80/14.76%	44.76/18.02%	68.01/19.23%	43.30
4.5	26.51/8.79%	33.81/14.64%	45.75/17.98%	52.97/18.29%	39.76
5.0	29.11/9.16%	36.16/15.02%	43.96/18.77%	63.66/21.33%	43.22
5.5	27.30/9.20%	36.53/15.15%	44.09/18.11%	60.35/18.96%	42.07
6.0	29.82/9.58%	41.54/16.12%	45.99/18.71%	52.75/18.48%	42.52
6.5	22.80/8.82%	33.87/15.01%	47.18/18.43%	70.78/19.82%	43.66
7.0	27.14/9.19%	35.76/15.19%	46.24/18.53%	67.92/19.85%	44.27
7.5	24.04/8.88%	34.79/14.91%	44.70/18.10%	66.88/19.48%	42.60
8.0	26.15/9.98%	37.37/16.02%	46.57/18.93%	67.36/19.76%	44.36

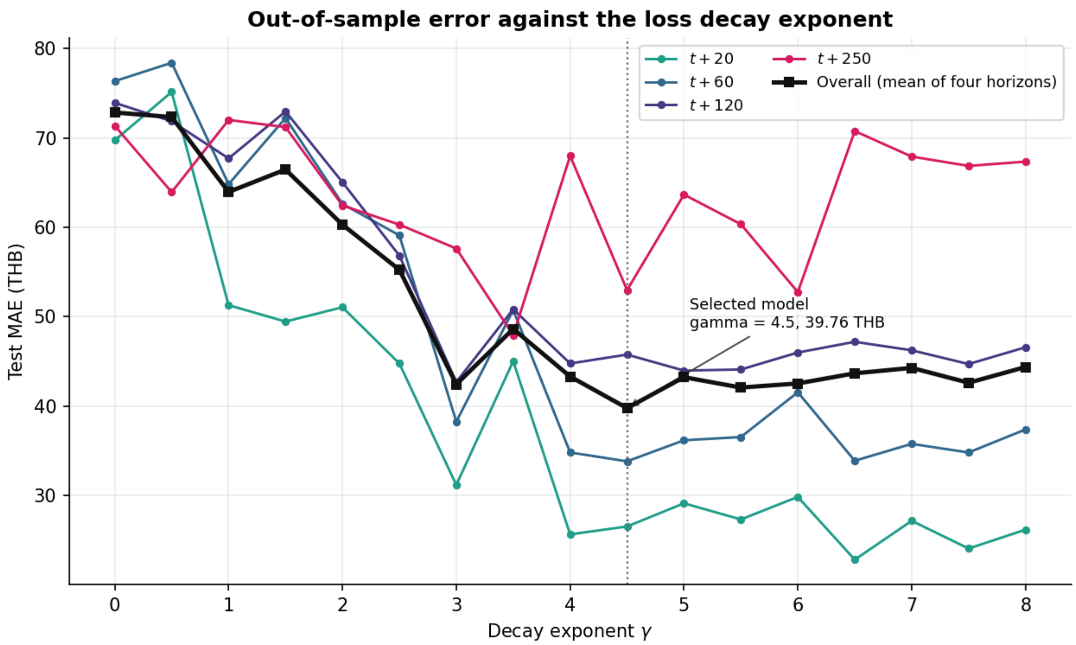

Figure S6: Horizon-specific and overall test MAE across the exponent grid.

S9 Qualitative forecasts and model diagnostics

The first trajectory panel deliberately includes contrasting cases, while the second uses one median-representative product per group. These single origins are diagnostic examples, not substitutes for aggregate evaluation.

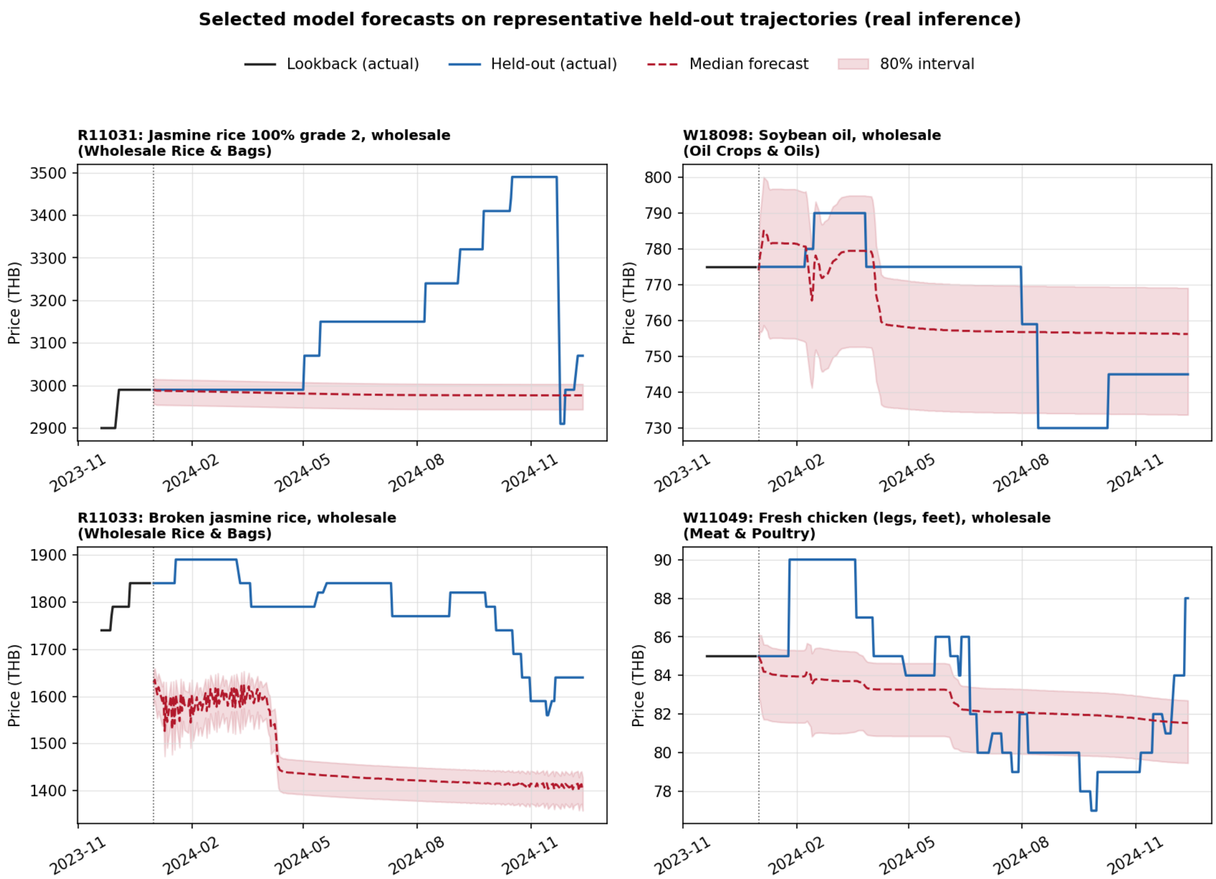

Figure S7: Contrasting held-out trajectories from the earliest 2024 origin.

Attention and variable-selection weights describe this fitted network, not causal economic effects. The oldest encoder position receives the largest averaged attention weight; observed price dominates historical selection; group and product identifiers dominate the informative static covariates; and day of week dominates known-future selection. Encoder length is constant in this experiment, so its static-selection weight is an architectural artifact.

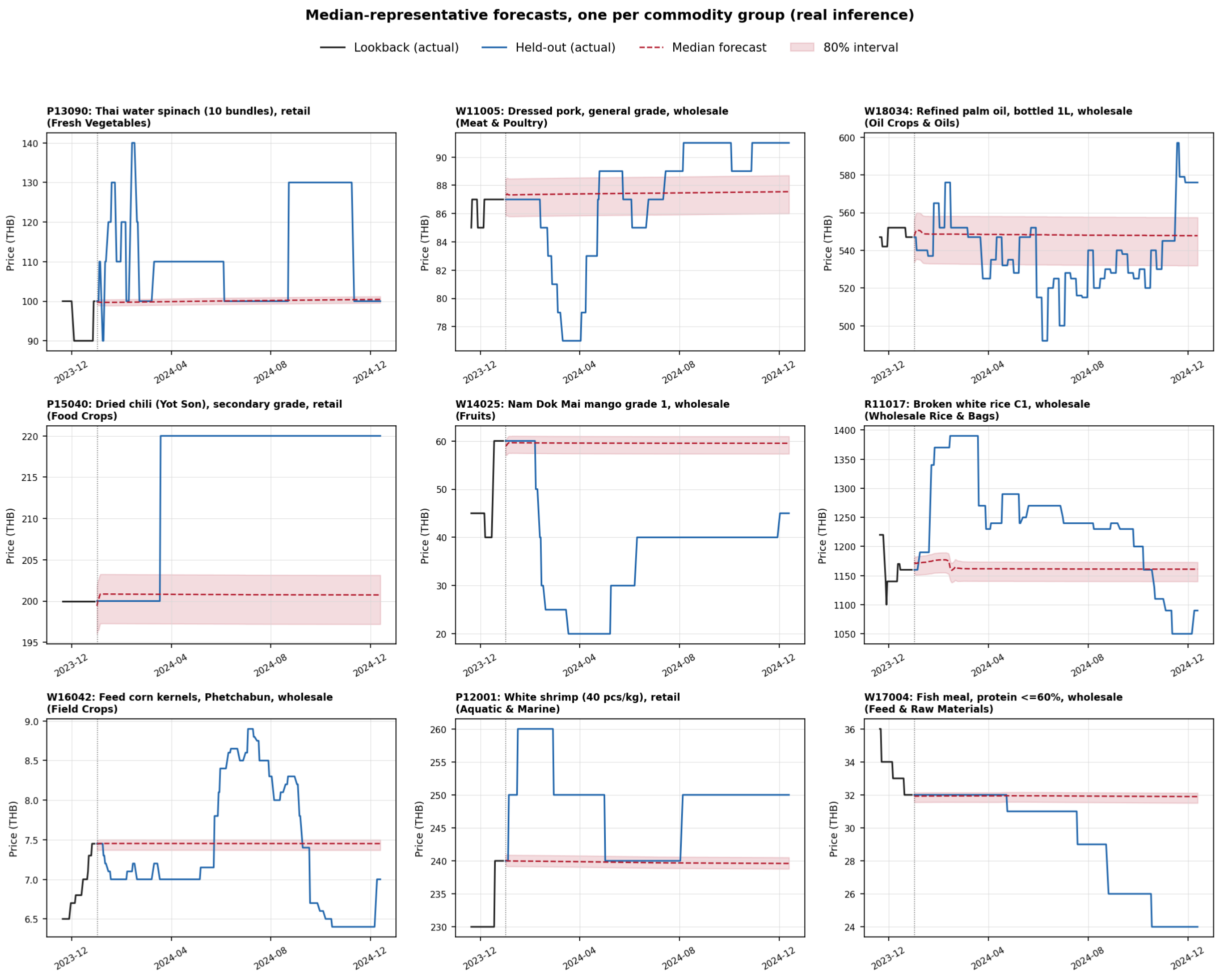

Figure S8: One median-representative held-out trajectory per commodity group.

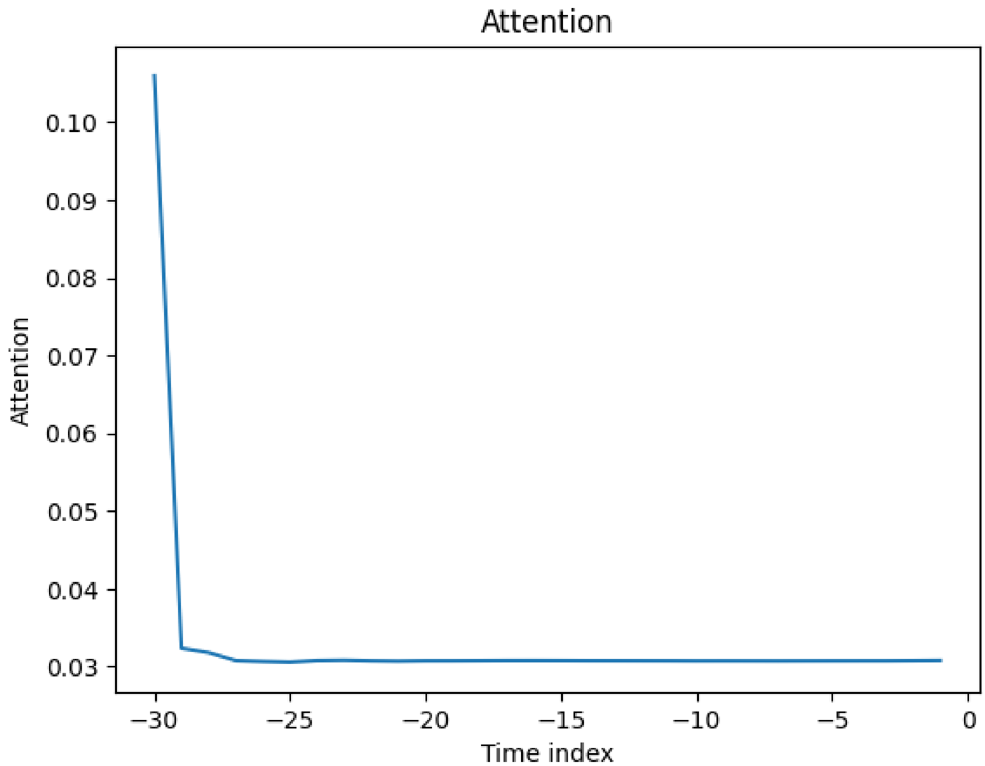

Figure S9: Averaged temporal attention over the 30-step encoder.

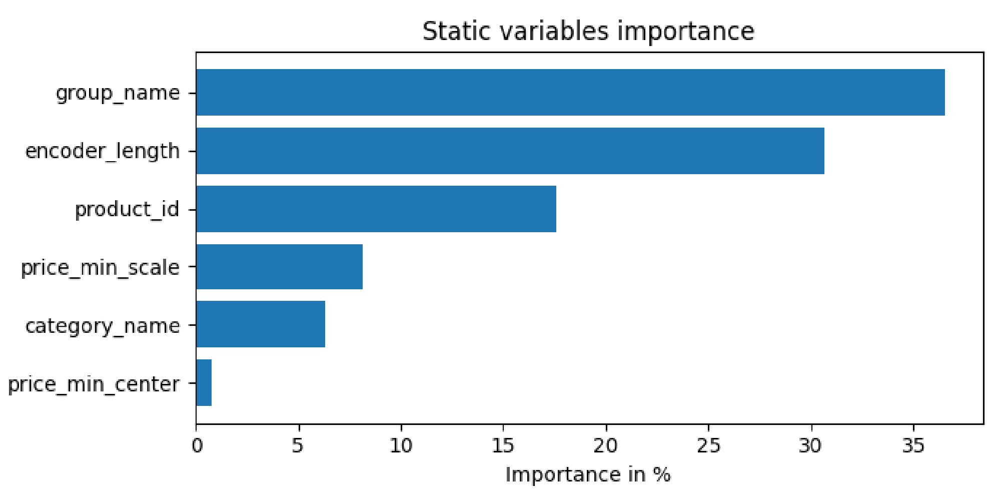

Figure S10: Static variable-selection weights.

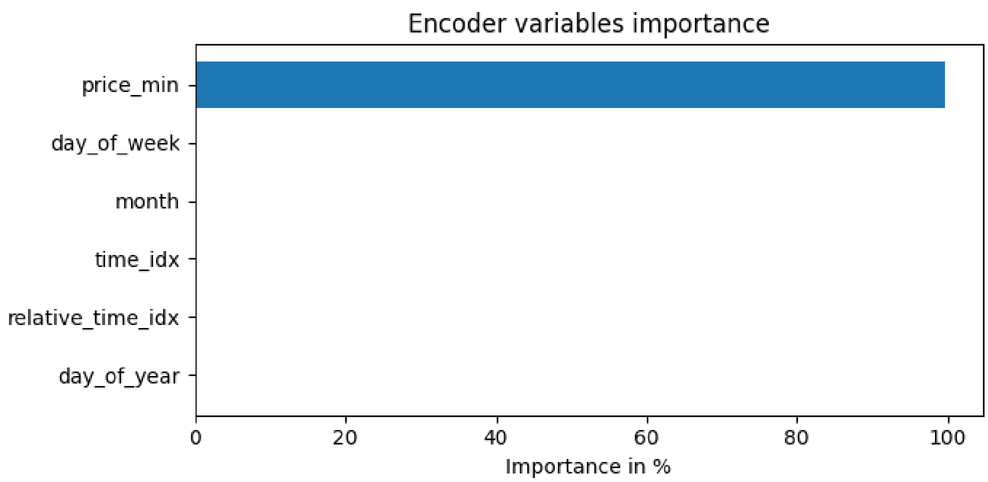

Figure S11: Historical encoder variable-selection weights.

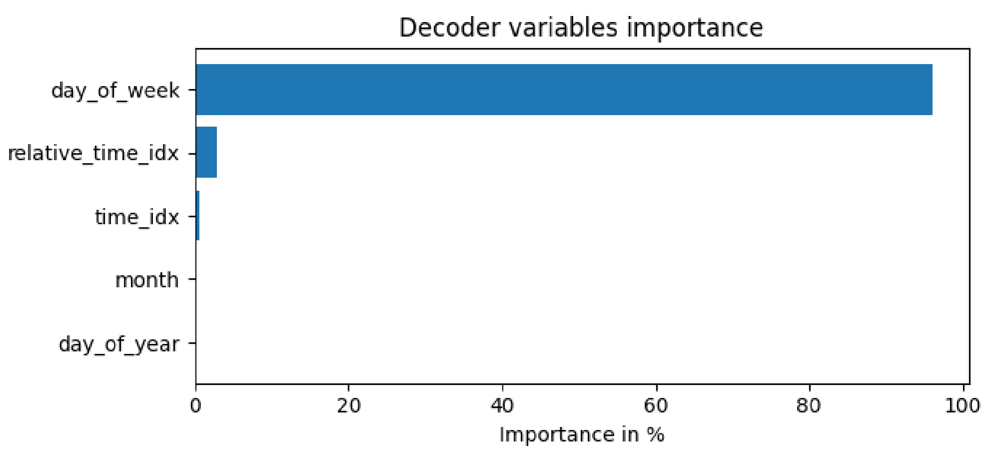

Figure S12: Known-future decoder variable-selection weights.

S10 Target-normalizer level-shift diagnostic

For 401 products, define the raw-price proxy

$$\mathrm{gap}_p = 100\,\frac{y_{p,\mathrm{last}}^{2023} - \bar{y}_p^{2018:2022}}{\bar{y}_p^{2018:2022}}.$$

This proxy is not the transformed internal normalizer center. Its absolute value exceeds 10% for 283 products (70.6%), with mean 30.4%, median 18.4%, and maximum 332%. Absolute gap correlates with relative selected-model error at $t{+}20$ (Spearman $r = 0.146$, unadjusted $p = 0.007$), reverses sign at $t{+}120$ ($r = -0.206$, $p < 0.001$) and $t{+}250$ ($r = -0.188$, $p < 0.001$), and does not identify a causal normalizer effect. The per-product inputs are released in `normalizer_staleness_check.csv` and `gap_vs_performance.csv`.

S11 ARIMA order

The local statistical reference uses ARIMA$(1, 1, 0)$, chosen a priori from the lag structure documented in Section 3.3 of the main paper (a unit root with a single partial-autocorrelation spike at lag one). We did not run a per-product order search for two reasons. First, Table 5 of the main paper shows the fitted ARIMA$(1, 1, 0)$ is already numerically indistinguishable from the persistence baseline at every horizon (differences of at most 0.01 THB), and the closed-form forecast function in Section 4.3 of the main paper explains why any low-order specification on a near-unit-root series must collapse onto that baseline. Second, the persistence baseline itself is retained as a separate reference, so the comparison against the strongest local statistical rule is already made directly.

S12 Summary

Under a matched 10-configuration validation-based architecture-search budget per class, Global LightGBM moves from 44.83 (original configuration, rerun) to 45.55 THB (+1.6%); Global MLP moves from 51.38 (original configuration, rerun) to 48.02 THB (-6.5%); Global LSTM moves from 54.58 (original configuration, rerun) to 54.61 THB (+0.1%); Global Transformer moves from 52.88 (original configuration, rerun) to 57.27 THB (+8.3%). No validation-selected reference configuration approaches the selected TFT of the main paper (39.76 THB overall), and none outperforms the matched persistence baseline overall. The conclusions of the main paper are therefore insensitive to reference-model tuning at this architecture-search budget, and the configurations reported in Table 5 give each reference class a representative rather than handicapped operating point.